

Seminar Quiz 2

The Economy

Sciences Po Paris - Reims

15 minutes. No extra papers. You are free to use calculators.

Exercise 1

1. What are the two key assumptions of the Malthusian model? Explain briefly using an analogy like we did in class. (2 points)

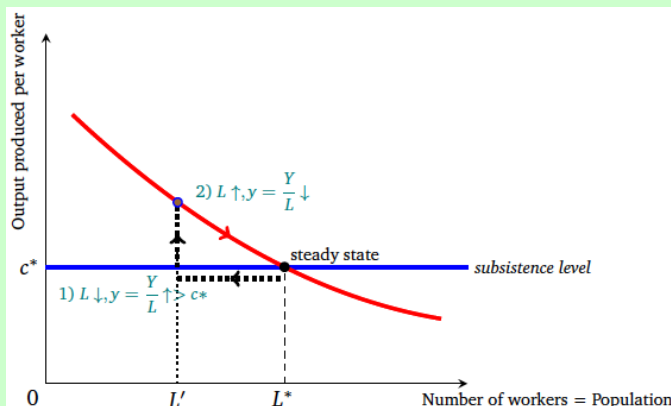
1. decreasing returns to scale for labor:

- as the number of farmers L , increases by 1 unit, the output per farmer $y = \frac{Y}{L}$ increases by less than 1 unit

2. population expands if living standards (wages) increase:

- if wage above subsistence level ($y > c^*$), then the population of farmers increase $L \uparrow$
- if wage below subsistence level ($y < c^*$), then the population of farmers decrease $L \downarrow$

2. Draw a graph of the Malthusian model. Show what would happen if there were a sudden disease which would result in a drop in the population. Keep your comments brief. (3 points)



- if L were to suddenly fall below L^* , then for fixed land, each farmer has more land = we can get a higher wage (y increases) (fact 1), but this drives up the birth rate, so $L \rightarrow L^*$ (fact 2).

Exercise 2

Consider an economy where people save 20% of their income, there is \$200 of capital per worker, and this capital depreciates by 30% in a given period t . Assume the following for the production function of this economy:

$$A = 1.5$$

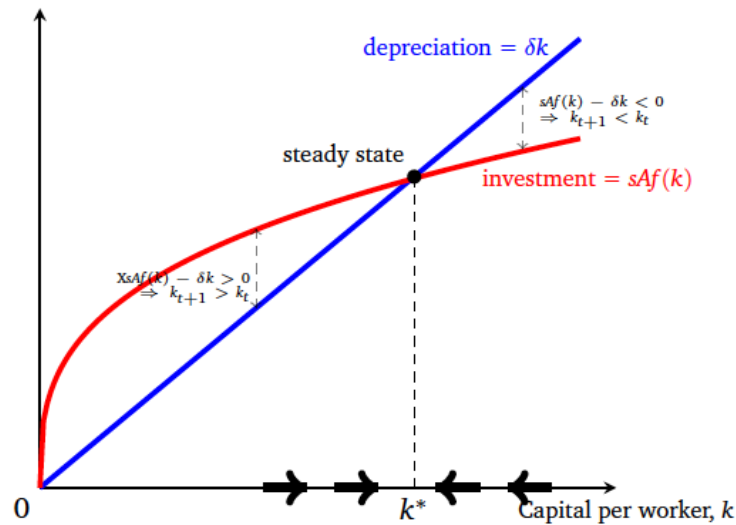
$$f(k) = f\left(\frac{k}{2}\right)$$

$$k_{t+1} = k_t - \delta k_t + sAf(k_t)$$

1. Given the capital accumulation formula above, calculate the **net change in capital** from period t to $t + 1$. (3 points)

$$k_{t+1} = 200 - \underbrace{0.3 * 200}_{\delta k_t} + \underbrace{0.2 * 1.5 * \frac{200}{2}}_{i_t} = \$170$$

$$k_{t+1} - k_t = -\$30$$



2. Recall the Solow model graph above. Which of the following is true for the aforementioned economy for the period t ? (2 points)

A) The economy is on the left part of the graph, relative to the steady state ($k_t < k^*$).

- B) The economy is on the right part of the graph, relative to the steady state ($k_t > k^*$).
- C) The economy is at the steady state ($k_t = k^*$).